

Day 21 : Top 5 packages and Libraries

1. Introduction to Packages and GetX Framework

Flutter provides a rich ecosystem of packages that help developers build applications faster and more efficiently. Packages reduce development time by providing ready-made solutions for common functionalities such as state management, navigation, local storage, networking, and utility features. In this project, various popular Flutter packages were explored and implemented.

2. GetX Package

GetX is a lightweight and powerful Flutter package used for state management, dependency injection, and route management.

Features

- State Management
- Route Management
- Dependency Injection
- High Performance
- Less Boilerplate Code

Installation

Add the following dependency in pubspec.yaml:

```
dependencies:  
get: ^4.7.2
```

Run:

```
flutter pub get
```

State Management with GetX

GetX allows reactive state management using .obs.

Example:

```
var count = 0.obs;
```

Navigation with GetX

Navigation can be performed without context.

```
Get.to(HomeScreen());
```

When to Use GetX

- Managing application state
- Navigation between screens
- Dependency management

3. Freezed Package

Freezed is used to generate immutable data classes and union types.

Features

- Immutable classes
- CopyWith method generation
- JSON serialization support
- Union/Sealed classes

Installation

dependencies:

freezed_annotation: ^2.4.1

json_annotation: ^4.9.0

dev_dependencies:

build_runner: ^2.4.13

freezed: ^2.5.7 json_serializable: ^6.8.0

Advantages

- Reduces boilerplate code.
- Makes code safer and easier to maintain.
- Useful for API response models.

4. Go Router Package

Go Router is an official Flutter package for declarative routing.

Features

- Declarative routing
- Deep linking support
- Nested navigation
- Typed routes

Installation :**dependencies:**

go_router: ^14.2.8

Advantages

- Simplifies navigation.
- Supports web URL navigation.

5. Dio Package

Dio is an advanced HTTP client for Flutter used for network requests.

Features

- API requests
- Request/Response interceptors
- File downloads
- Request cancellation
- Timeout configuration

Installation :**dependencies:**

dio: ^5.7.0

Example

```
Dio dio = Dio();
```

```
var response = await dio.get(url);
```

6. Hive Package

Hive is a lightweight and fast NoSQL database for Flutter.

Features

- Local storage
- Fast performance
- No native dependencies
- Lightweight

Installation :

dependencies:

hive: ^2.2.3

hive_flutter: ^1.1.0

Box Concept

Hive stores data inside containers called Boxes.

Example:

```
var box = Hive.box('myBox');
```

CRUD Operations

1. Create

```
box.put('name', 'Sneha');
```

2. Read

```
box.get('name');
```

3. Update

```
box.put('name', 'John');
```

4. Delete

```
box.delete('name');
```

7. Utility Packages

1. intl :

Used for:

- Date formatting
- Currency formatting
- Internationalization

Example:

```
DateFormat('dd-MM-yyyy').format(DateTime.now());
```

2. url_launcher

Used to open:

- Websites
- Emails
- Phone calls

Example :

```
launchUrl(Uri.parse('https://flutter.dev'));
```

3. share_plus

Used to share text, files, and images.

Example:

```
Share.share('Hello Flutter');
```

4. connectivity_plus

Used to check internet connectivity status.

Example:

```
Connectivity().checkConnectivity();
```

Application Implementation

A Flutter application was developed using multiple packages.

Packages Used:

1. GetX - State Management
2. Dio - API Integration
3. Hive - Local Database Storage
4. connectivity_plus - Internet Status

Application Features:

- Fetch news data using Dio API.
- Manage state using GetX.
- Store favourite news using Hive.
- Check network connectivity using connectivity_plus.

OUTPUT :

